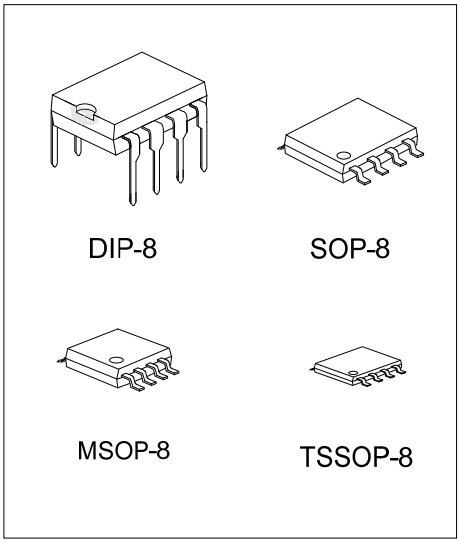
DUAL OPERATIONAL AMPLIFIER



■ DESCRIPTION

The UTC **M2904** consists of two independent, high gain, internally frequency compensated operation amplifiers which were designed specifically to operate from a single power supplies is also possible and the low power supply current drain is independent of the magnitude of the power supply voltage.

■ FEATURES

*Single Supply
*Operating Voltage: +3V ~ +32V
*Low Operating Current: 0.7mA (typ.)
*Slew Rate: 0.5V/μs (typ.)

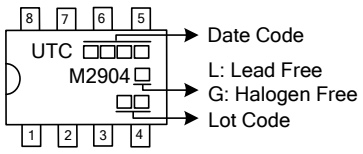
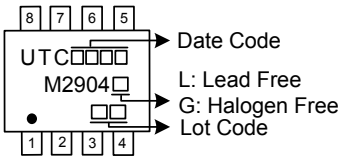
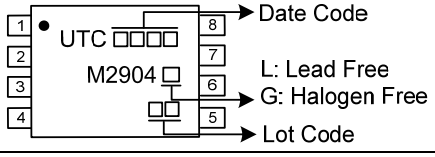
■ ORDERING INFORMATION

Order Number		Package	Packing
Lead Free	Halogen Free		
M2904L-D08-T	M2904G-D08-T	DIP-8	Tube
M2904L-S08-R	M2904G-S08-R	SOP-8	Tape Reel
M2904L-SM1-R	M2904G-SM1-R	MSOP-8	Tape Reel
M2904L-P08-R	M2904G-P08-R	TSSOP-8	Tape Reel

M2904G-D08-T
(1)Packing Type
(2)Package Type
(3)Green Package

(1) T: Tube, R: Tape Reel
(2) D08: DIP-8, S08: SOP-8, SM1: MSOP-8, P08: TSSOP-8
(3) G: Halogen Free and Lead Free, L: Lead Free

■ MARKING

PACKAGE	MARKING
DIP-8	
SOP-8 / MSOP-8	
TSSOP-8	

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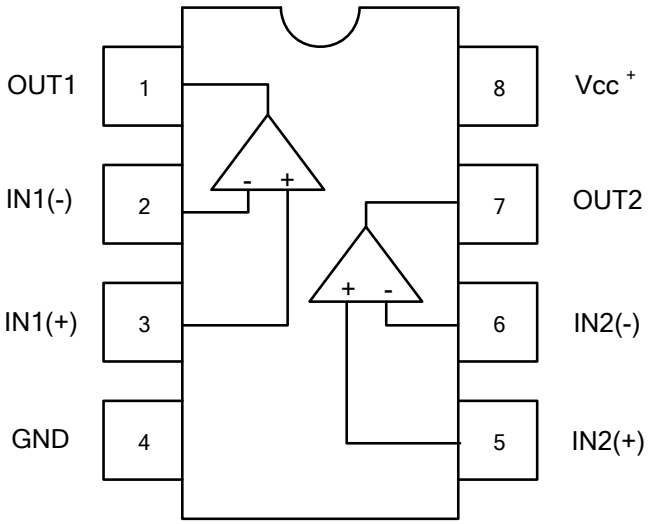
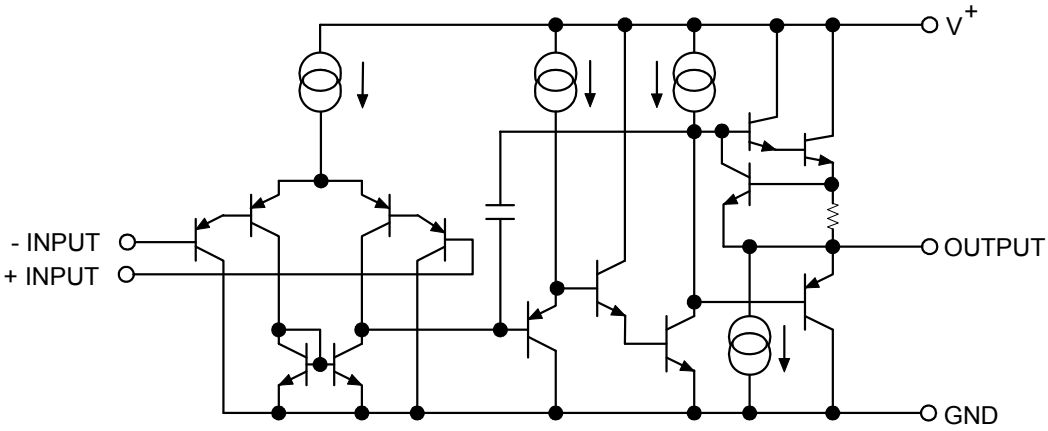
■ PIN CONFIGURATION



■ EQUIVALENT CIRCUIT (1/2 shown)



■ ABSOLUTE MAXIMUM RATINGS (T_A=25°C, unless otherwise specified)

PARAMETER		SYMBOL	RATINGS	UNIT
Supply Voltage		$V^+(V^+/V^-)$	32(or±16)	V
Differential Input Voltage		$V_{I(DIFF)}$	±32	V
Input Voltage		V_{IN}	-0.3 ~ +32	V
Power Dissipation	DIP-8	P_D	625	mW
	SOP-8		450	mW
	TSSOP-8		360	mW
	MSOP-8		300	mW
Junction Temperature		T_J	+150	°C
Operating Temperature		T_{OPR}	-40 ~ +125	°C
Storage Temperature		T_{STG}	-40 ~ +150	°C

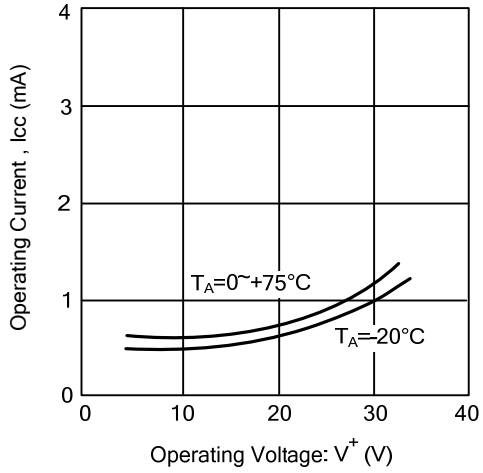
Note: Absolute maximum ratings are those values beyond which the device could be permanently damaged. Absolute maximum ratings are stress ratings only and functional device operation is not implied.

■ ELECTRICAL CHARACTERISTICS (V+=5V, T_A=25°C, unless otherwise specified)

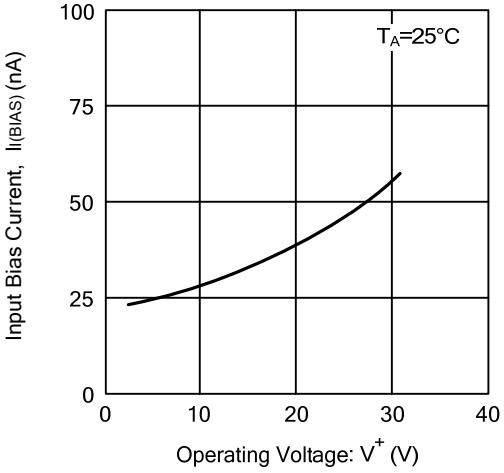
PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Input Offset Voltage	$V_{I(OFF)}$	R_S=0Ω		2	7	mV
Input Offset Current	$I_{I(OFF)}$			5	50	nA
Input Bias Current	$I_{I(BIAS)}$			25	250	nA
Large Signal Voltage Gain	Gv	$R_L \geqq$ 2kΩ		100		dB
Maximum Output Voltage Swing	V_{OM}	R_L=2kΩ	3.5			V
Input Common Mode Voltage	$V_{I(CM)}$		0~3.5			V
Common Mode Rejection Ratio	RR			85		dB
Supply Voltage Rejection Ratio	SVR			100		dB
Output Source Current	I_{SOURCE}	V_{IN}^+=1V,V_{IN}^-=0V	20	30		mA
Output Sink Current	I_{SINK}	V_{IN}^+=0V,V_{IN}^-=1V	8	20		mA
Channel Separation	CS	f=1k ~20kHz,Input Referred		120		dB
Operating Current	I_{CC}	R_L=∞		0.7	1.2	mA
Slew Rate	SR	V^+/V^-=±15V		0.5		V/μs
Unity Gain Bandwidth	f_T	V^+/V^-=±15V		0.2		MHz

■ TYPICAL CHARACTERISTICS

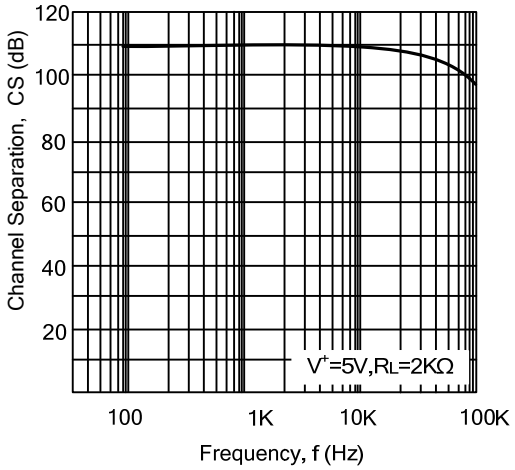
Operating Current vs.Operating Voltage



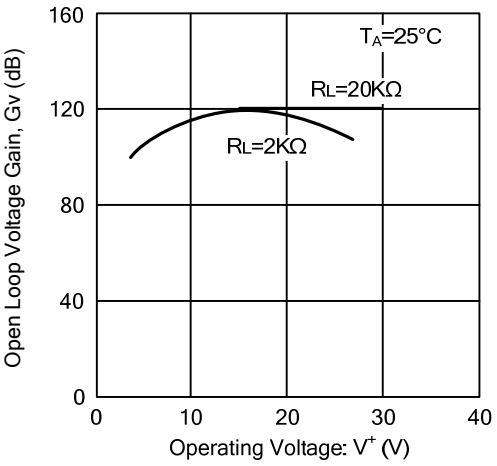
Input Bias Current vs.Operating Voltage



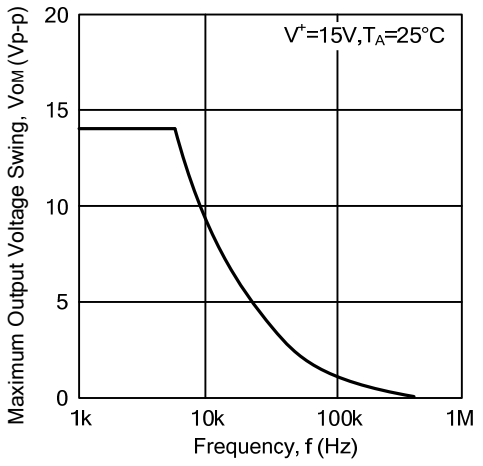
Channel Separation vs. Frequency



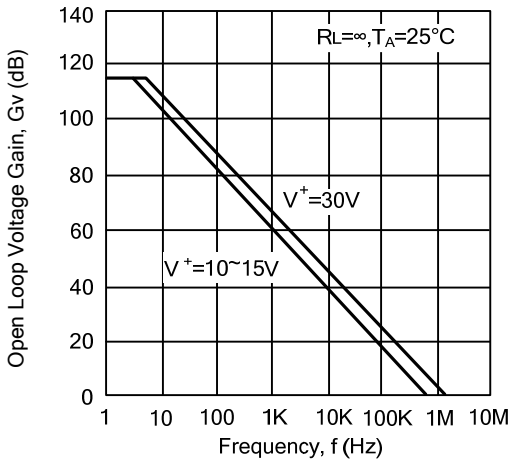
Voltage Gain vs.Operating Voltage



Maximum Output Voltage Swing vs.Frequency

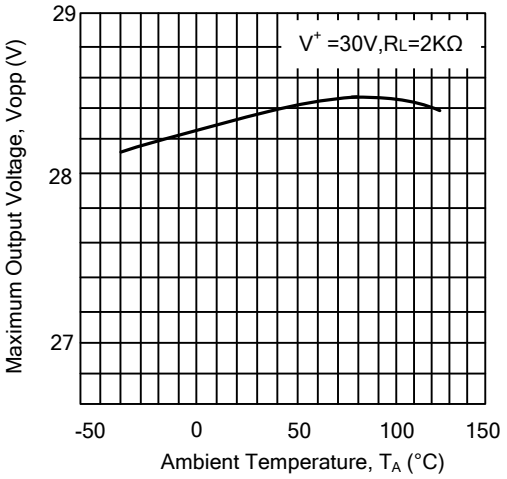


Open Loop Voltage Gain vs. Frequency

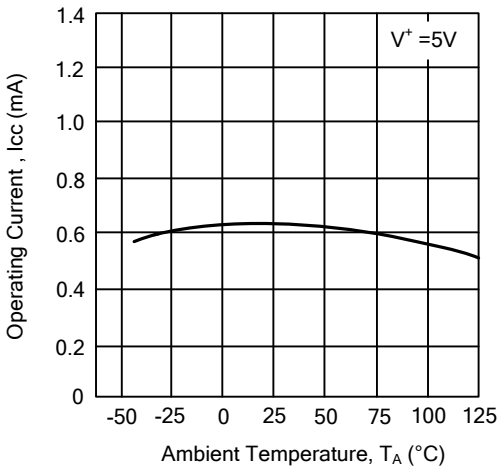


■ TYPICAL CHARACTERISTICS(Cont.)

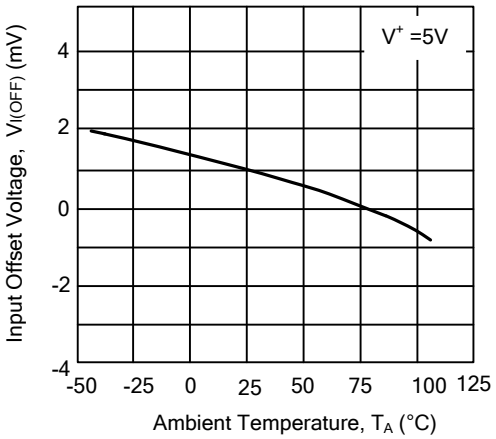
Maximum Output Voltage Swing vs. Temperature



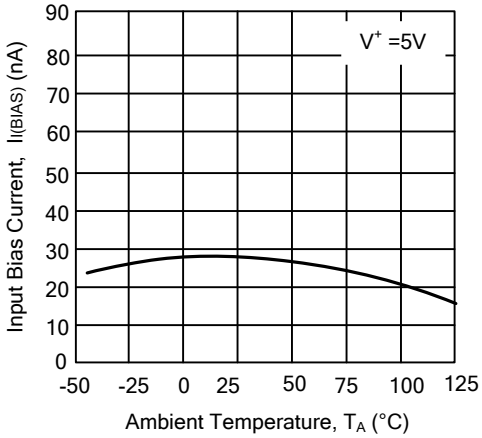
Operating Current vs.Temperature



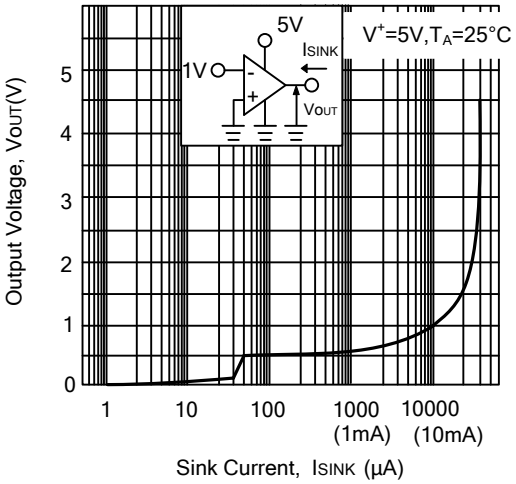
Input Offset Voltage vs.Temperature



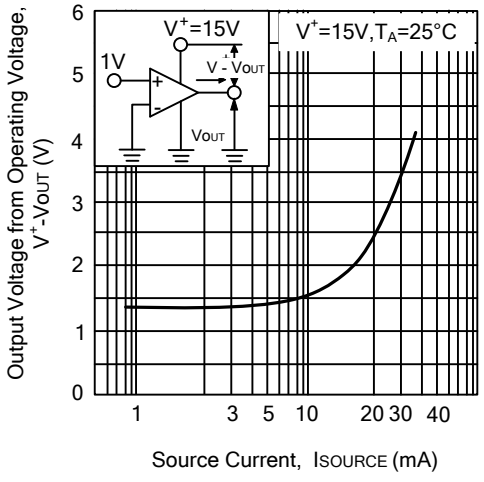
Input Bias Current vs.Temperature



Output Voltage vs.Sink Current

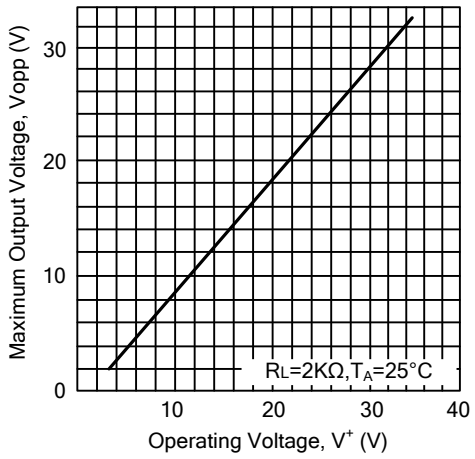


Source Current

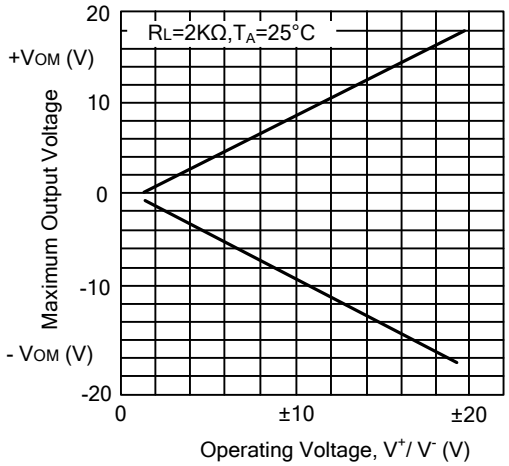


■ TYPICAL CHARACTERISTICS (Cont.)

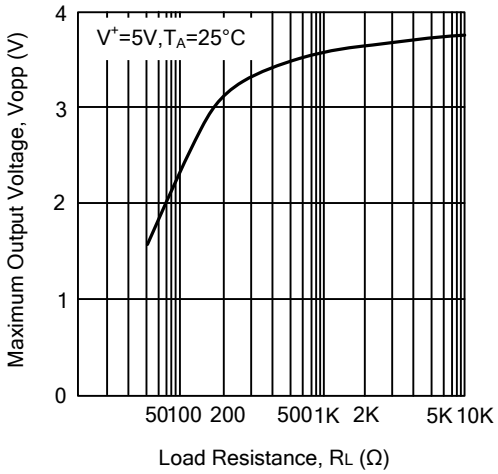
Maximum Output Voltage



Maximum Output Voltage vs.Operating Voltage



Maximum Output Voltage Swing vs..Load Resistance



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